Course Notes

Lecture 09 — 6LoWPAN: IPv6 in Networks of Constrained Devices

Communication Technologies for the Internet of Things

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Text to support independent study

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How to Use These Notes

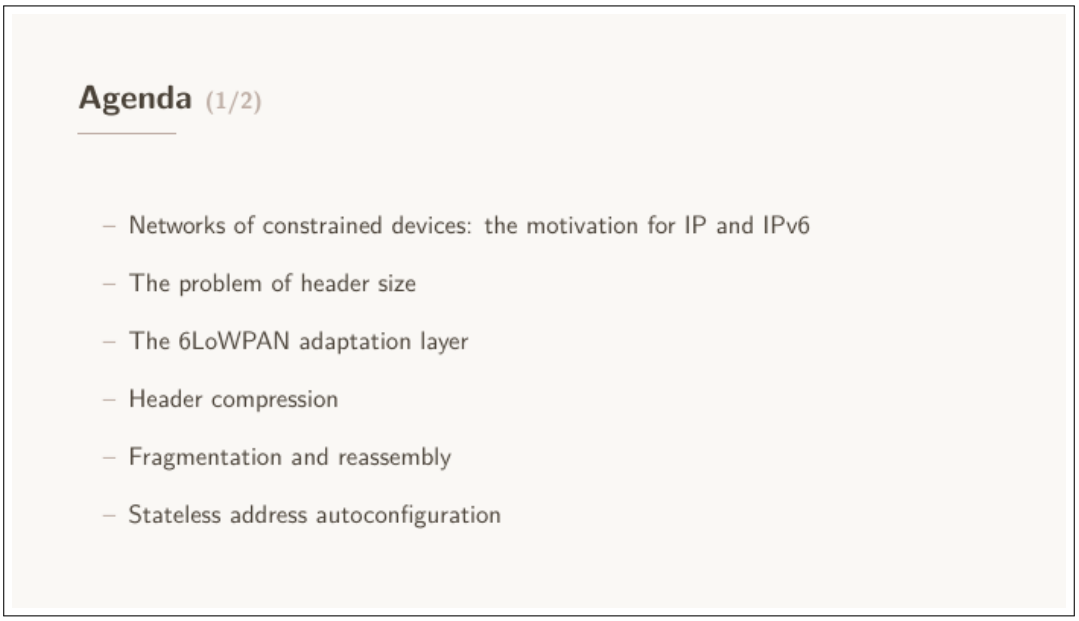
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

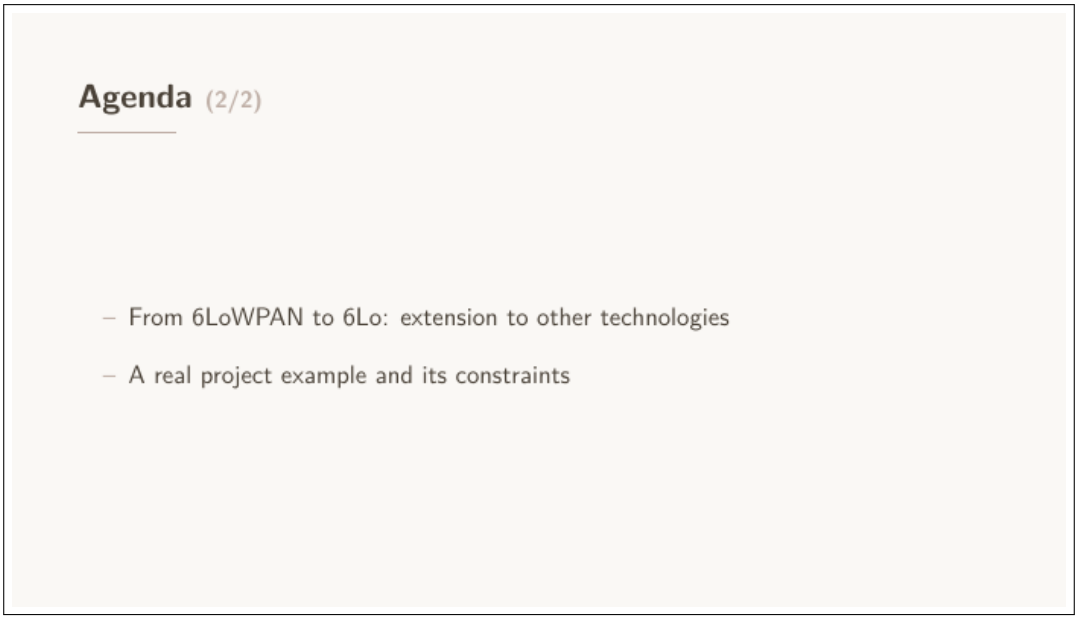
1. Commentary on the Slides

Slide 2. Agenda (1/2)



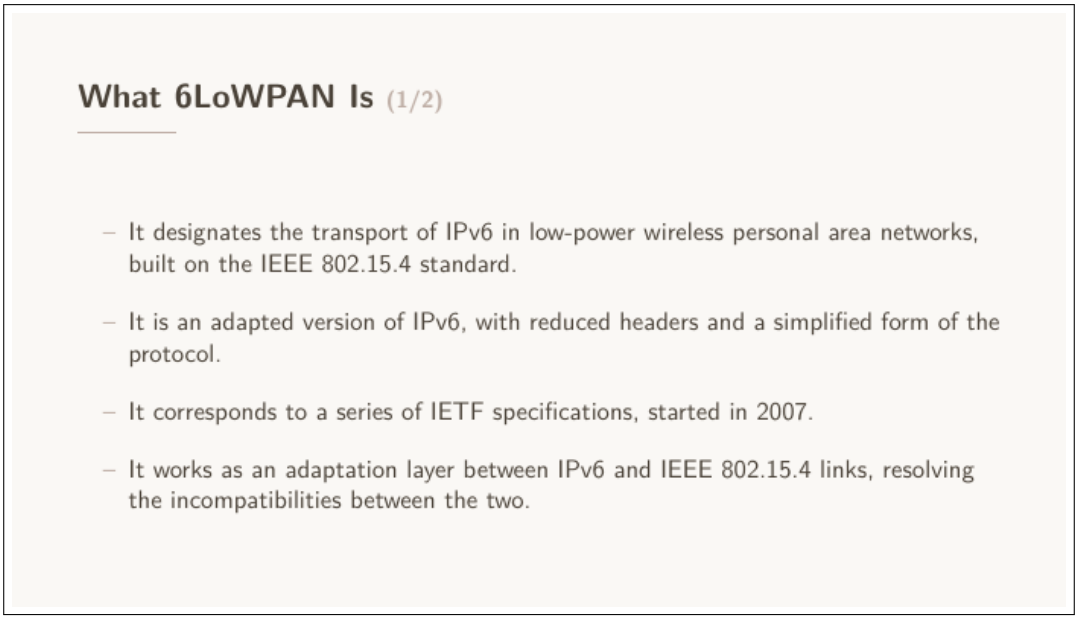
The common thread. The lecture starts from a concrete problem: how to make IPv6, designed for unconstrained networks, work on devices with tiny memory, energy and frames. The agenda follows the answer layer by layer: motivation, the header problem, the adaptation layer, compression, fragmentation and autoconfiguration.

Slide 3. Agenda (2/2)



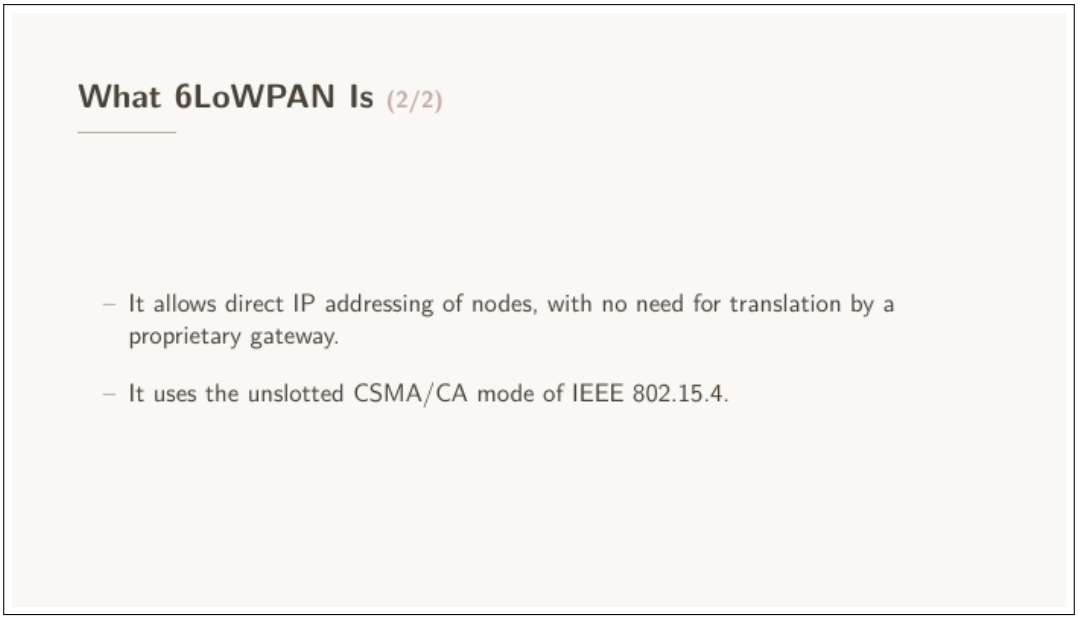
From the standard to the real case. The extension to 6Lo shows that the solution stopped being specific to IEEE 802.15.4. The case study closes the lecture with an engineering lesson: the decision not to use IP is also a design decision, and you should be able to justify it.

Slide 4. What 6LoWPAN Is (1/2)



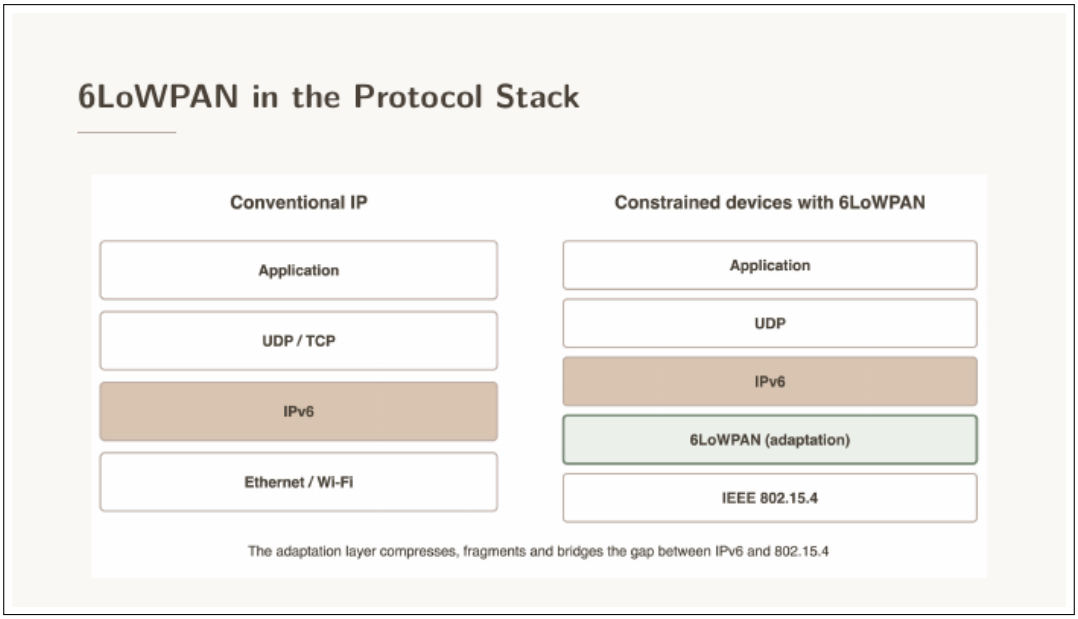
An adaptation layer. 6LoWPAN is not a new network protocol. It is a layer between IPv6 and the 802.15.4 link that resolves the incompatibilities between the two: frame size, layer-2 addresses and the cost of headers.

Slide 5. What 6LoWPAN Is (2/2)



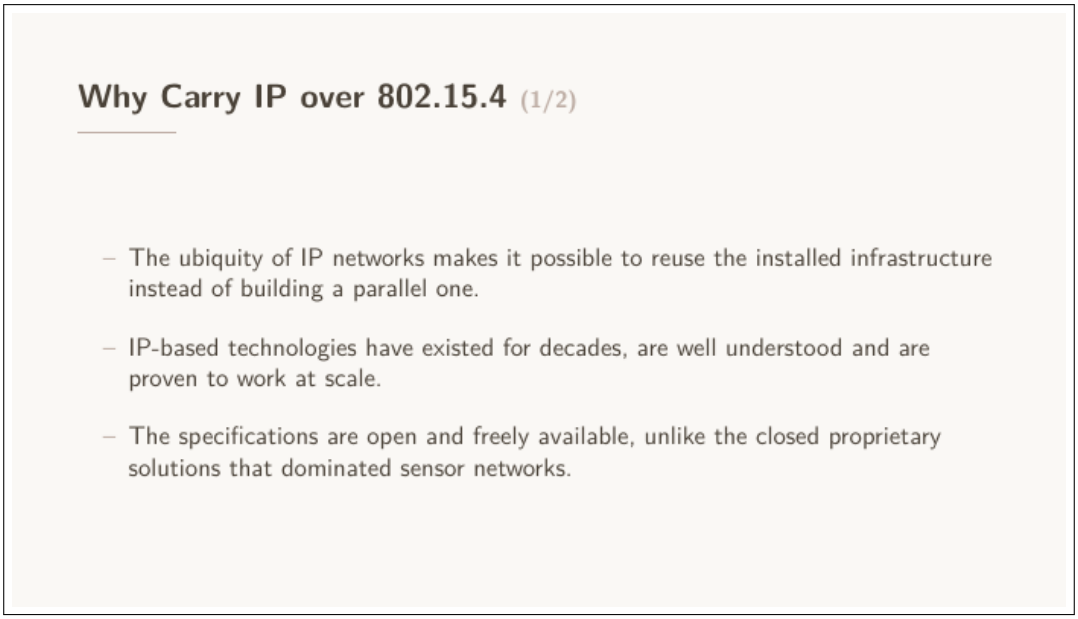
No proprietary gateway. The practical consequence is that each node can be addressed directly, without a gateway translating between a proprietary protocol and IP. Medium access is the unslotted CSMA/CA of 802.15.4, which shows that 6LoWPAN changes neither the physical layer nor medium access.

Slide 6. 6LoWPAN in the Protocol Stack



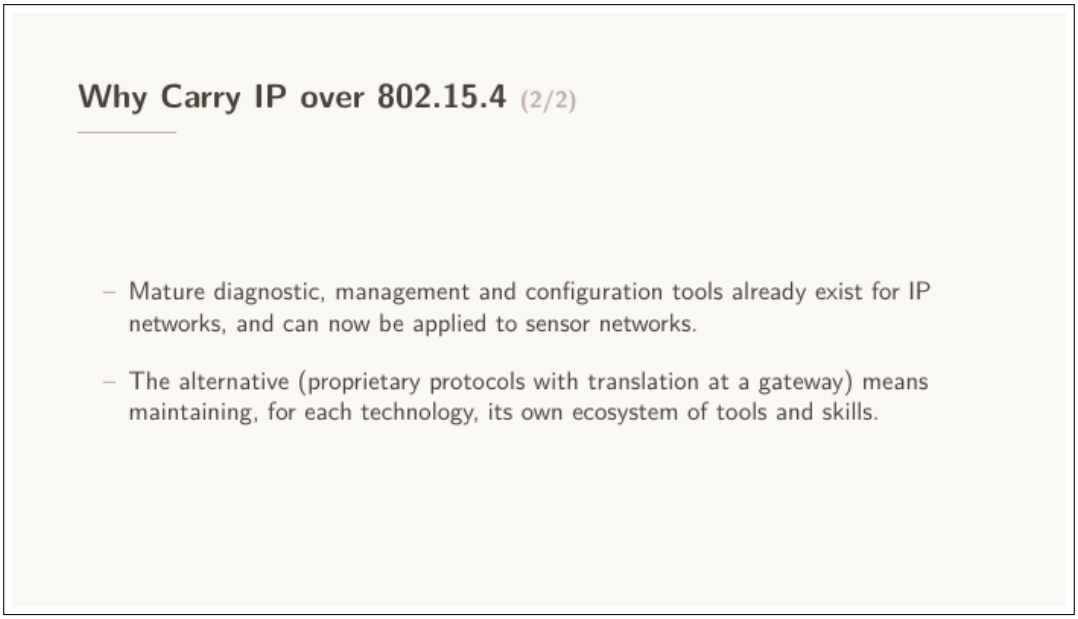
Where the layer sits. The figure compares the conventional stack with that of a constrained device. The difference is a single layer between IPv6 and 802.15.4, in green. Everything above, UDP and application, stays the same, which is why existing IP applications can be reused.

Slide 7. Why Carry IP over 802.15.4 (1/2)



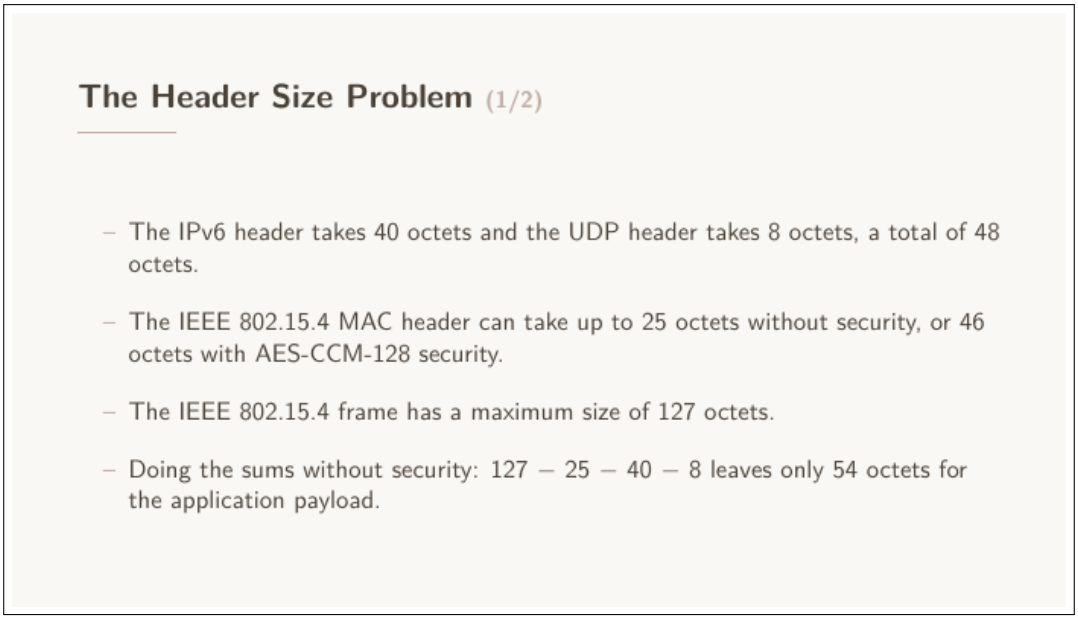
Reusing what exists. The three arguments are reused infrastructure, proven technology and open specifications. Together they explain why the industry abandoned proprietary protocols for sensor networks.

Slide 8. Why Carry IP over 802.15.4 (2/2)



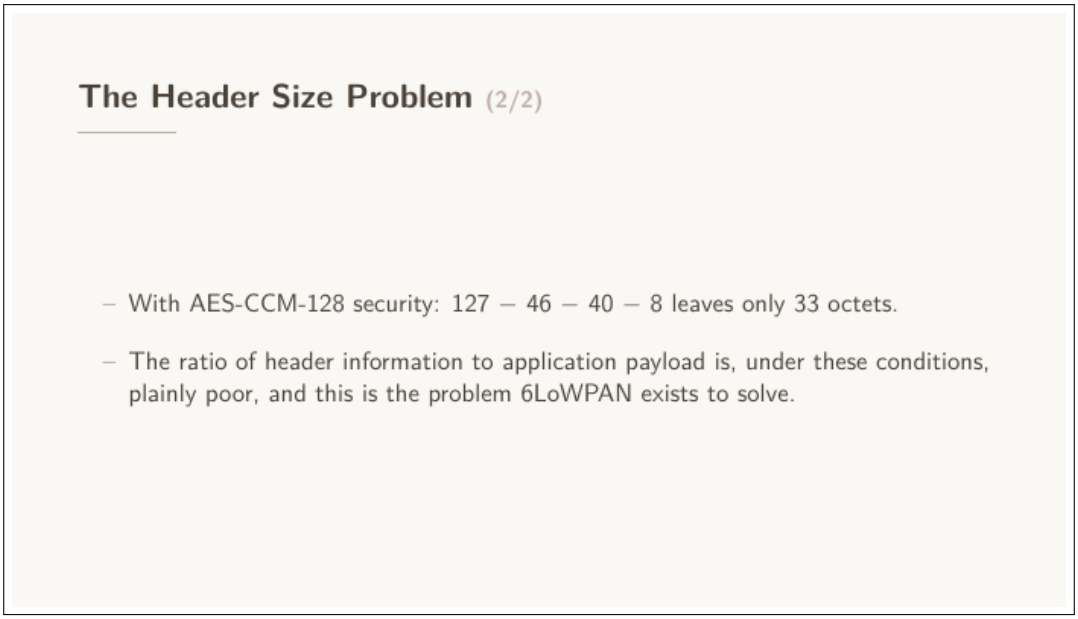
Tools and skills. The least obvious argument is tools: ping, traceroute, packet capture and management now serve a sensor network. With a proprietary gateway, each technology would need its own ecosystem of tools.

Slide 9. The Header Size Problem (1/2)



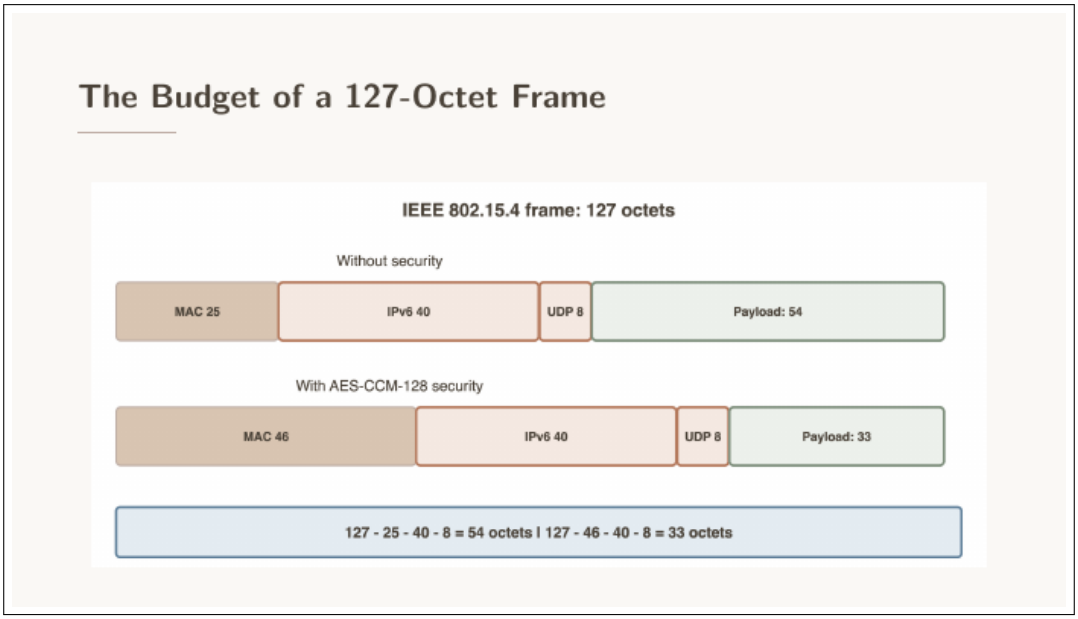
The sum that does not add up. An 802.15.4 frame has 127 octets. After the MAC header, IPv6 and UDP, 54 octets of data are left without security. The arithmetic is simple and is the reason for the whole lecture.

Slide 10. The Header Size Problem (2/2)



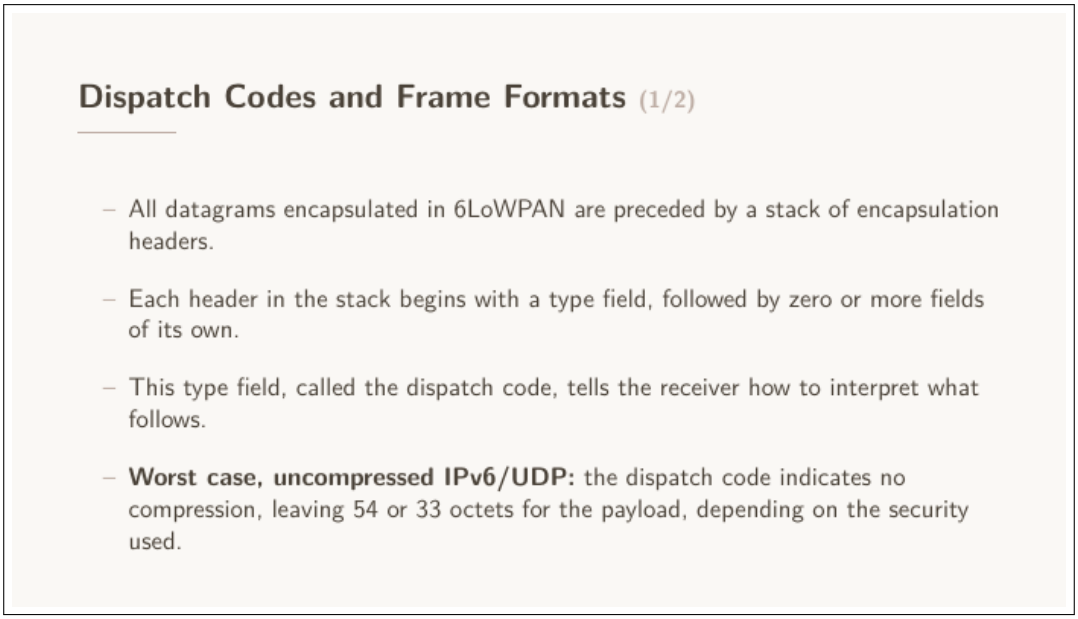
With security, even worse. With AES-CCM-128 encryption and authentication, the MAC header grows to 46 octets and only 33 are left. More than half of the frame is header. Check: can you redo both sums without looking at the slide?

Slide 11. The Budget of a 127-Octet Frame



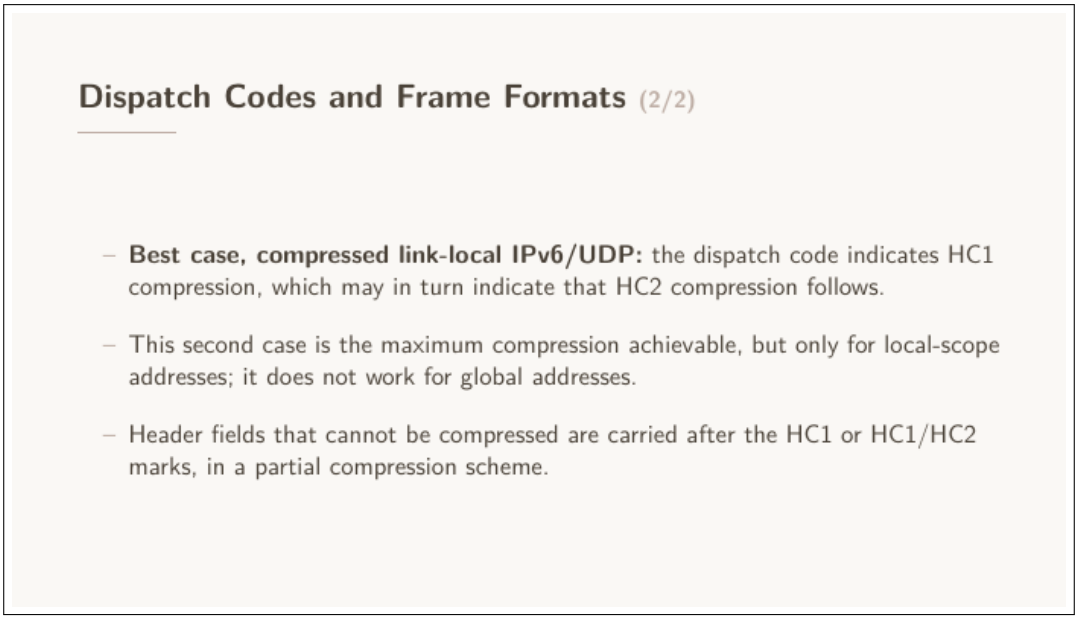
The budget as bars. The bars in the figure are proportional in width to the number of octets. The green area, the payload, is the one the application cares about, and it is plainly the smallest. Comparing the two bars shows the cost of security.

Slide 12. Dispatch Codes and Frame Formats (1/2)



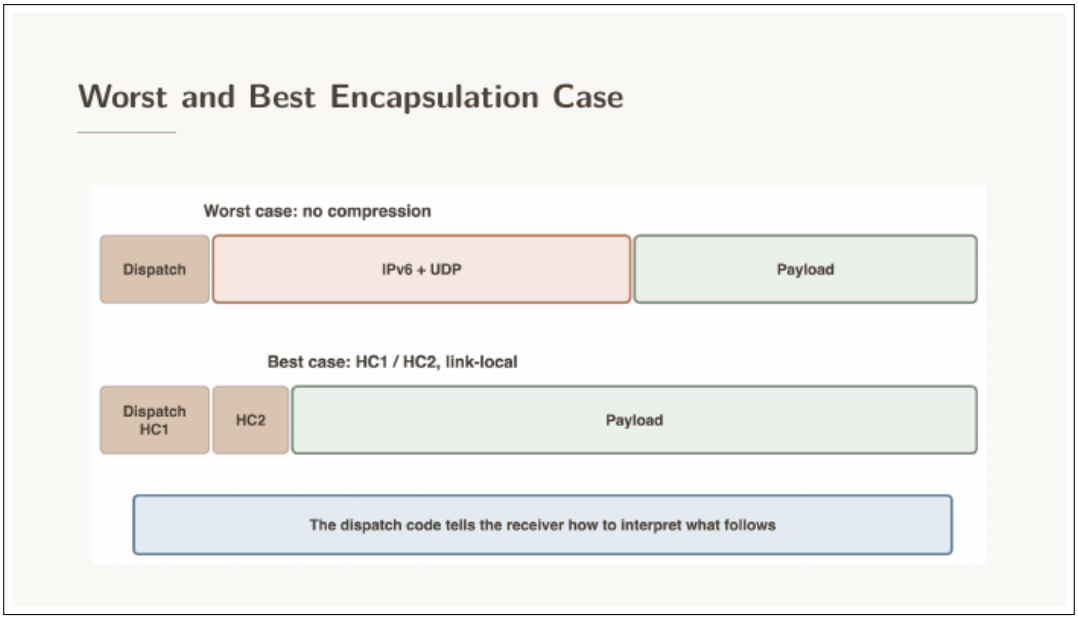
Dispatch: how to read what follows. The first field of each 6LoWPAN header, the dispatch code, tells the receiver how to interpret the rest. It is what lets compressed and uncompressed packets coexist on the same network.

Slide 13. Dispatch Codes and Frame Formats (2/2)



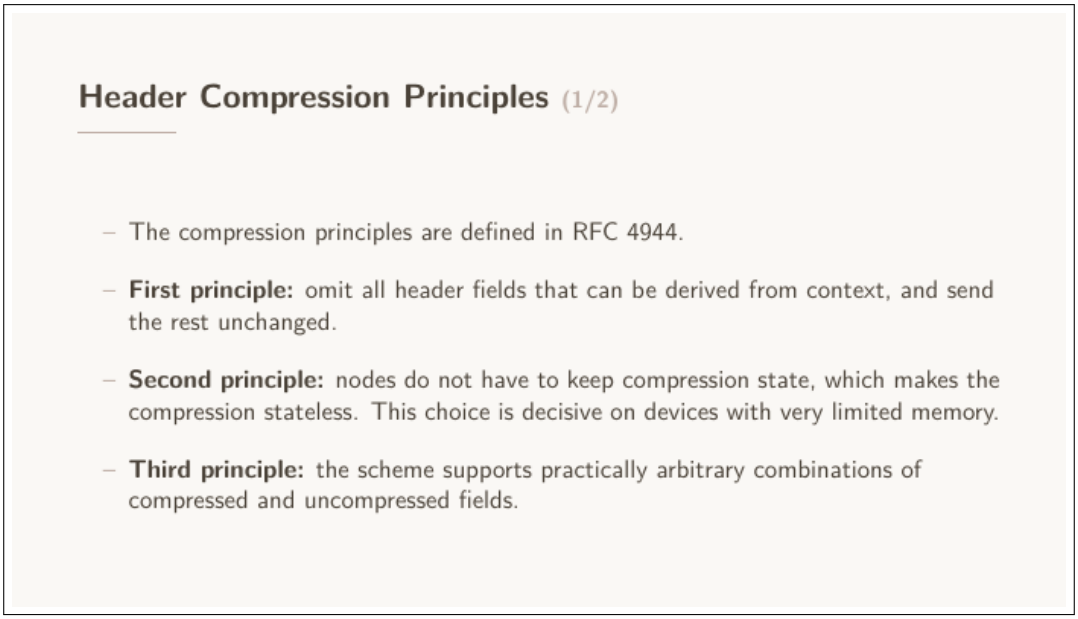
Maximum compression and its conditions. Maximum compression only works with link-local addresses. Outside that scope, global addresses have to be carried, and compression is partial. That is why compressing global addresses required more work.

Slide 14. Worst and Best Encapsulation Case



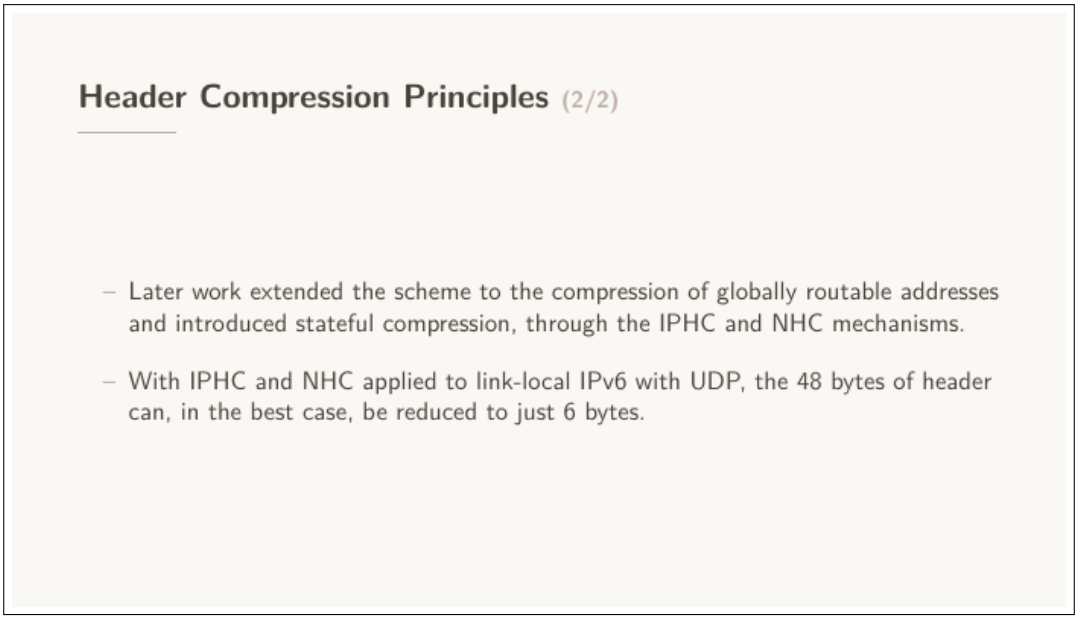
Worst and best case. The figure contrasts the two extremes. At the top, the uncompressed packet with 40 and 8 octets of header. Below, HC1 and HC2 reduce the header to one or two octets, freeing space for the payload.

Slide 15. Header Compression Principles (1/2)



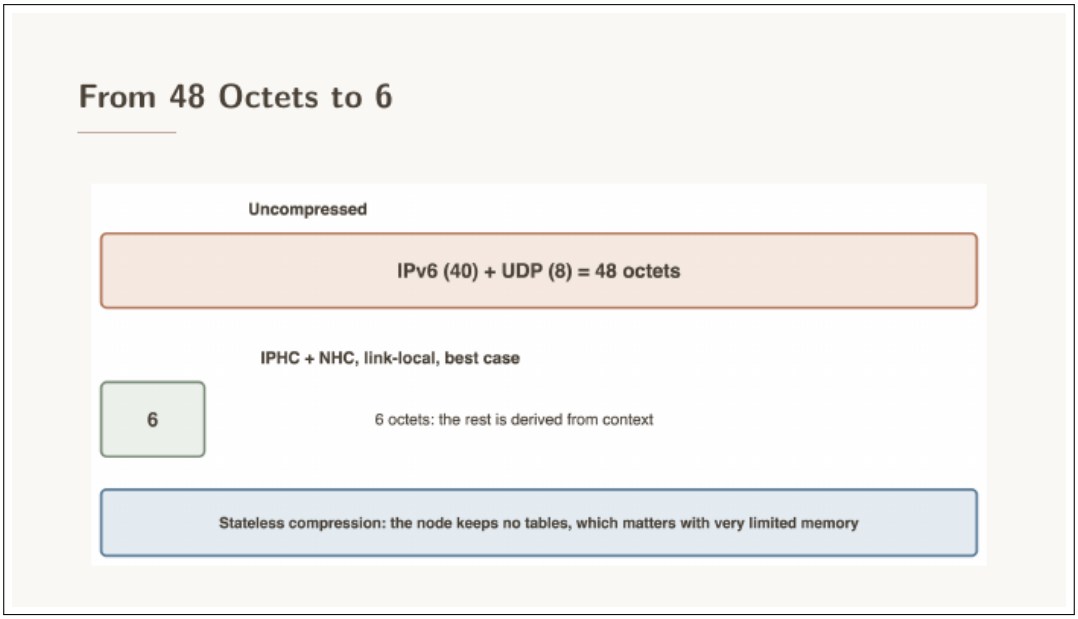
Three principles. Omit what can be derived from context, keep no state and allow arbitrary combinations of compressed fields. The second is the most important for constrained devices: with no state, there are no compression tables to keep in memory.

Slide 16. Header Compression Principles (2/2)



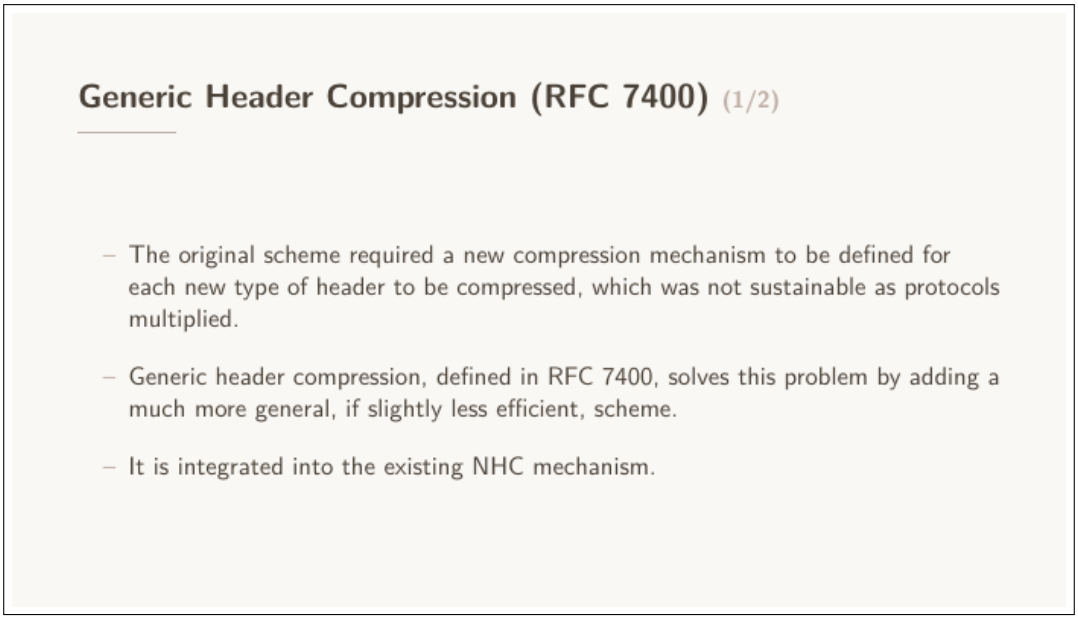
From HC1 to IPHC and NHC. The original standard was extended to compress global addresses and to allow stateful compression. In the best case, link-local IPv6 and UDP are reduced to 6 octets.

Slide 17. From 48 Octets to 6



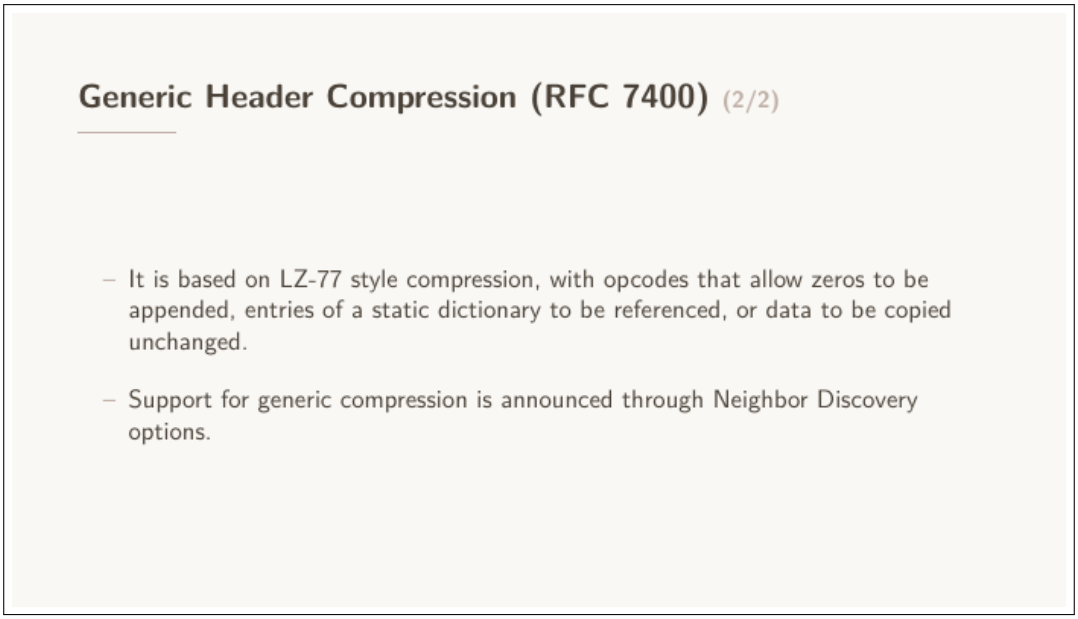
From 48 to 6 octets. The figure shows the gain: the 48 octets of IPv6 and UDP shrink to 6. The reason is that most fields can be inferred: the version is always 6, the length is already in the frame, and link-local addresses derive from MAC addresses.

Slide 18. Generic Header Compression (RFC 7400) (1/2)



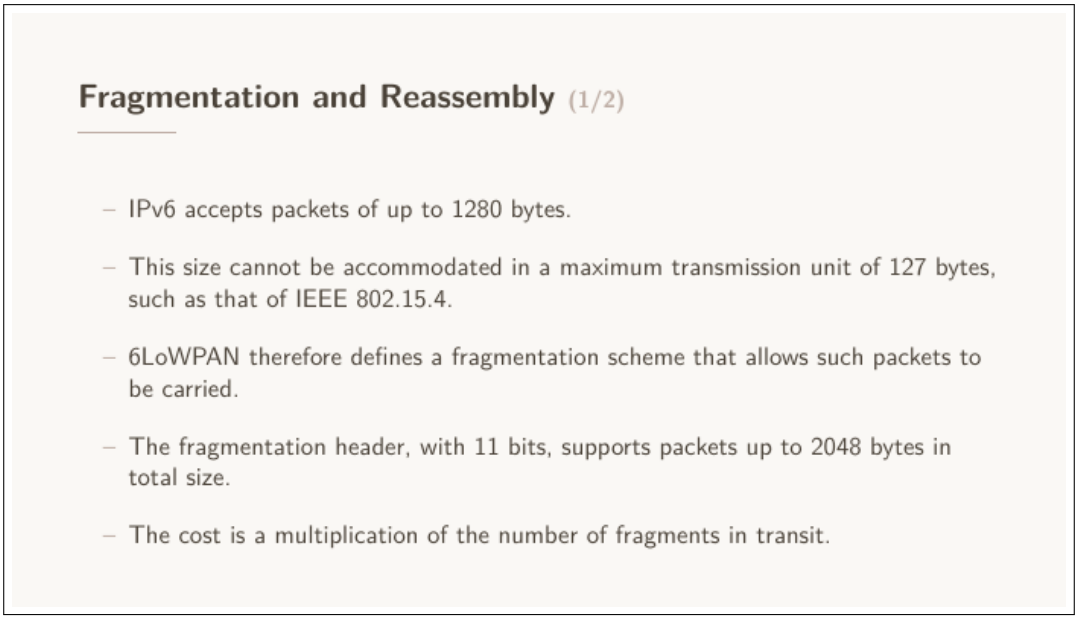
One mechanism per protocol. The original scheme required compression to be specified for each new header. That did not scale. The generic compression of RFC 7400 solves the problem with a single scheme, although less efficient than the specific ones.

Slide 19. Generic Header Compression (RFC 7400) (2/2)



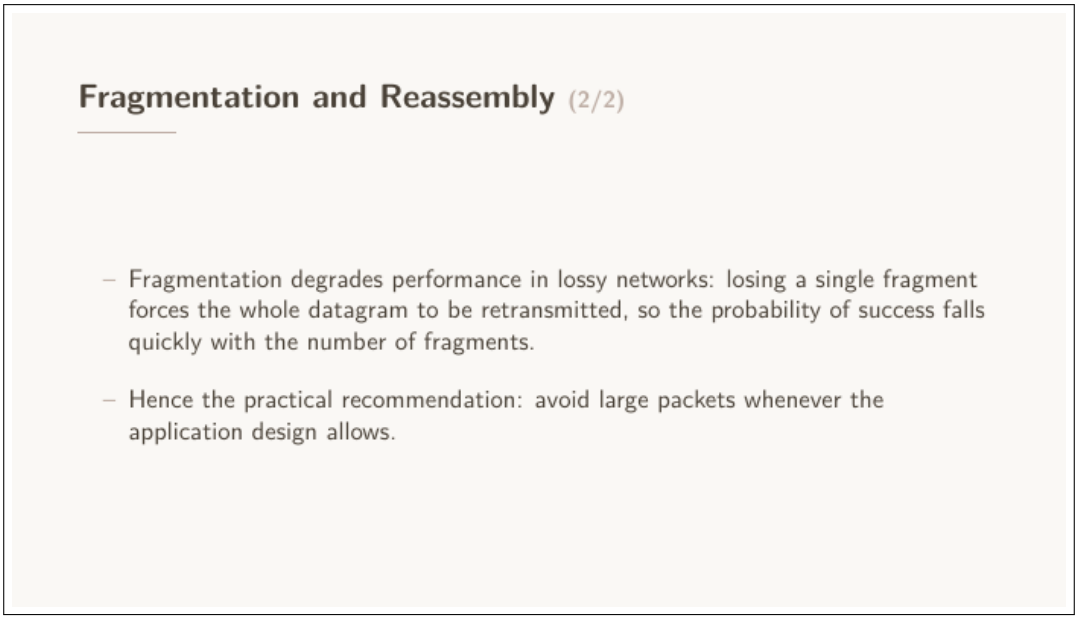
LZ-77 style. The mechanism copies, repeats and references a static dictionary, and support is announced through Neighbor Discovery options. For the student, the essential point is the idea: when there is no specific compression, there is a general one.

Slide 20. Fragmentation and Reassembly (1/2)



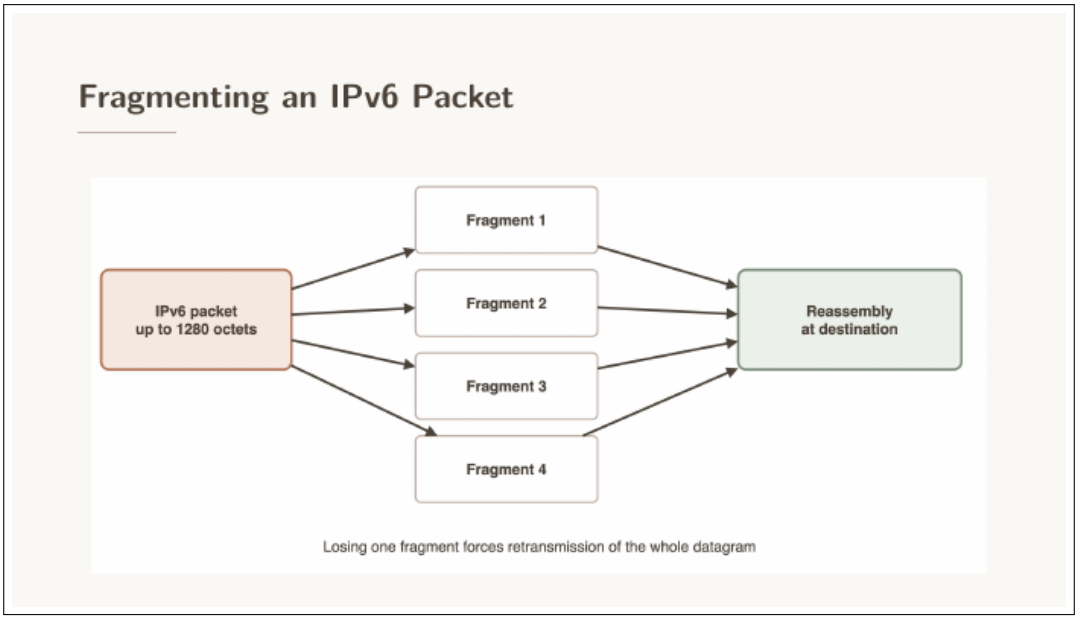
1280-octet packets. IPv6 requires the link to carry at least 1280 octets. The 802.15.4 frame has 127. The solution is to fragment, and 6LoWPAN defines its own fragmentation and reassembly mechanism.

Slide 21. Fragmentation and Reassembly (2/2)



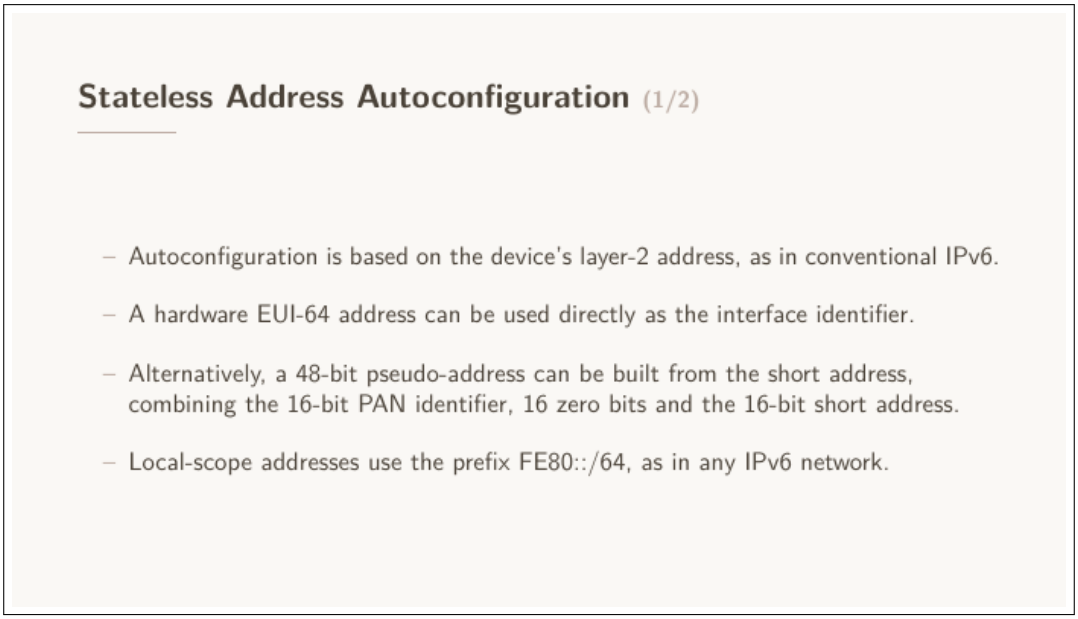
Fragmentation has a price. In lossy networks, each lost fragment forces the whole datagram to be retransmitted. The probability of delivery falls quickly with the number of fragments. The recommendation is one of design: keep messages small.

Slide 22. Fragmenting an IPv6 Packet



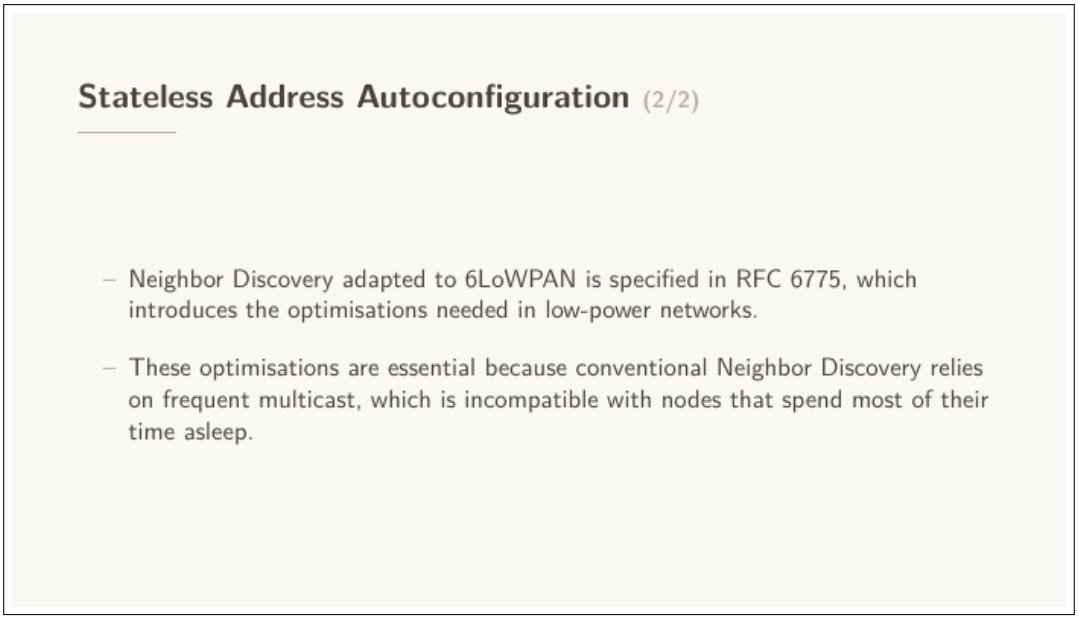
Fragmenting and reassembling. The packet is split into fragments that travel separately and are put back together at the destination. If one fragment fails, the destination discards everything and the sender retransmits. The figure summarises the two consequences: more frames and more risk.

Slide 23. Stateless Address Autoconfiguration (1/2)



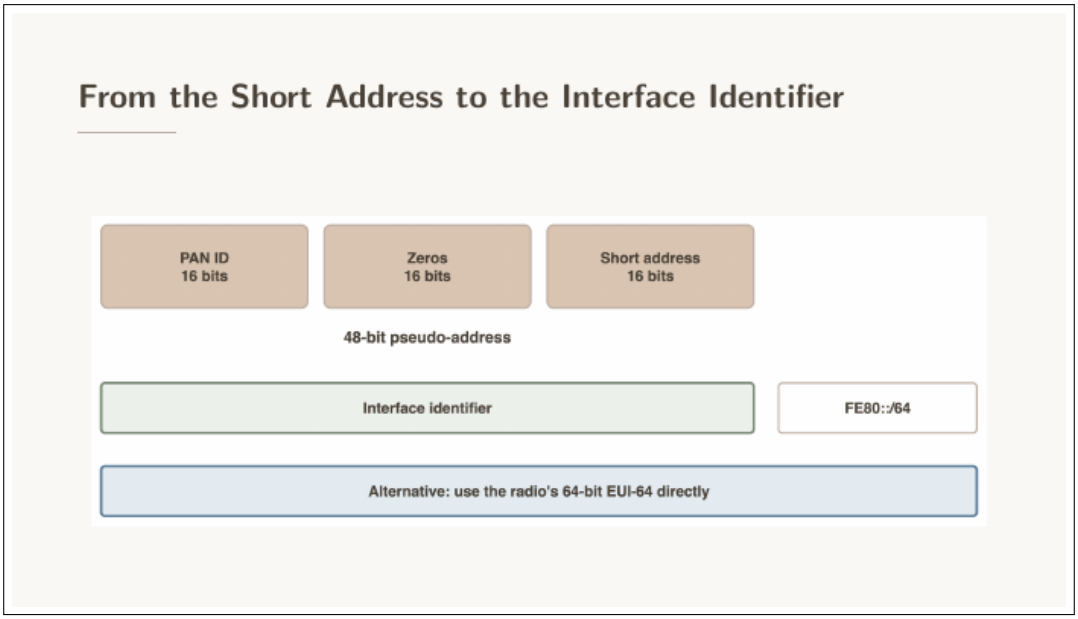
Addresses from hardware. As in IPv6, the interface identifier derives from the layer-2 address. The radio's EUI-64 can be used directly. When the device only has the 16-bit short address, a 48-bit pseudo-address is built.

Slide 24. Stateless Address Autoconfiguration (2/2)



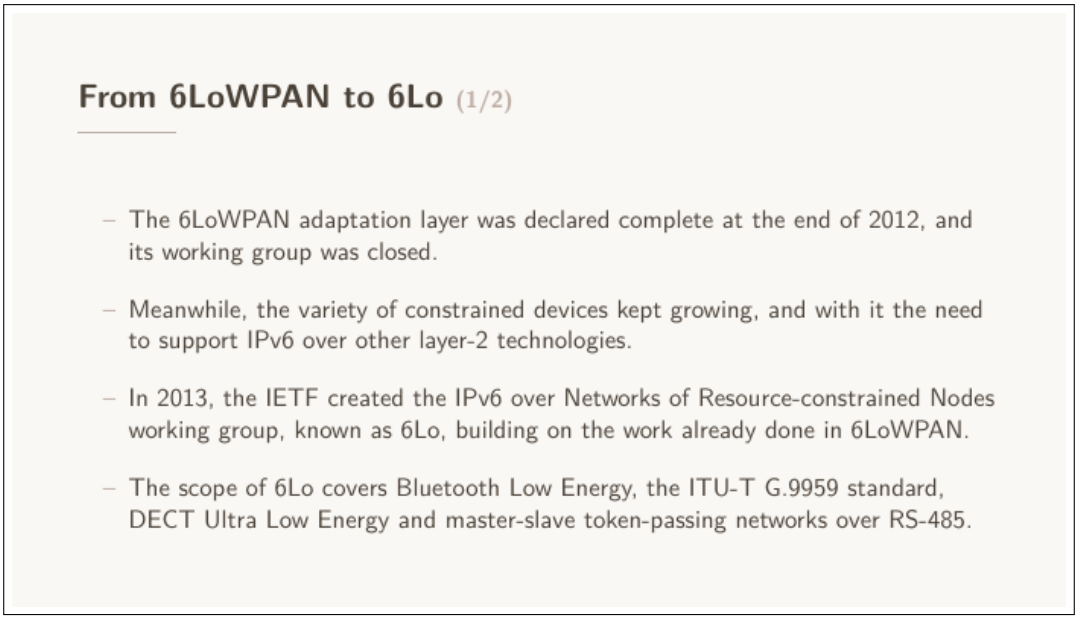
Optimised Neighbor Discovery. Conventional Neighbor Discovery depends on frequent multicast, which is incompatible with sleeping nodes. RFC 6775 introduces explicit address registration with the router and reduces the use of multicast, which is what makes discovery viable in low-power networks.

Slide 25. From the Short Address to the Interface Identifier



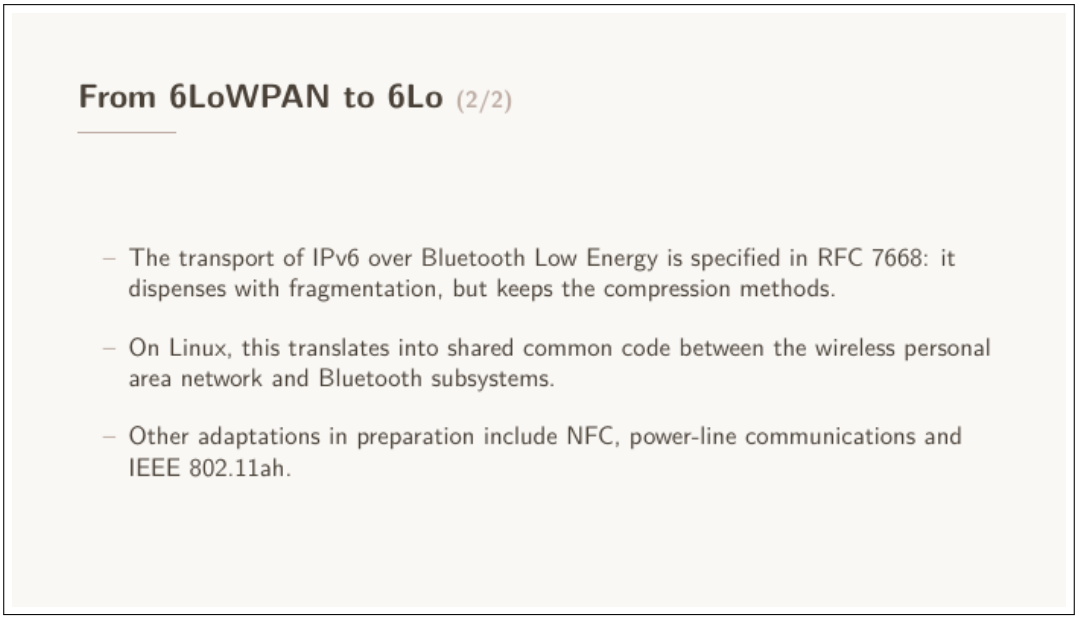
Building the identifier. The figure shows how the pseudo-address is built: PAN identifier, 16 zero bits and short address. Combined with the FE80::/64 prefix it gives the link-local address. Check: which part of the address identifies the network and which part identifies the node?

Slide 26. From 6LoWPAN to 6Lo (1/2)



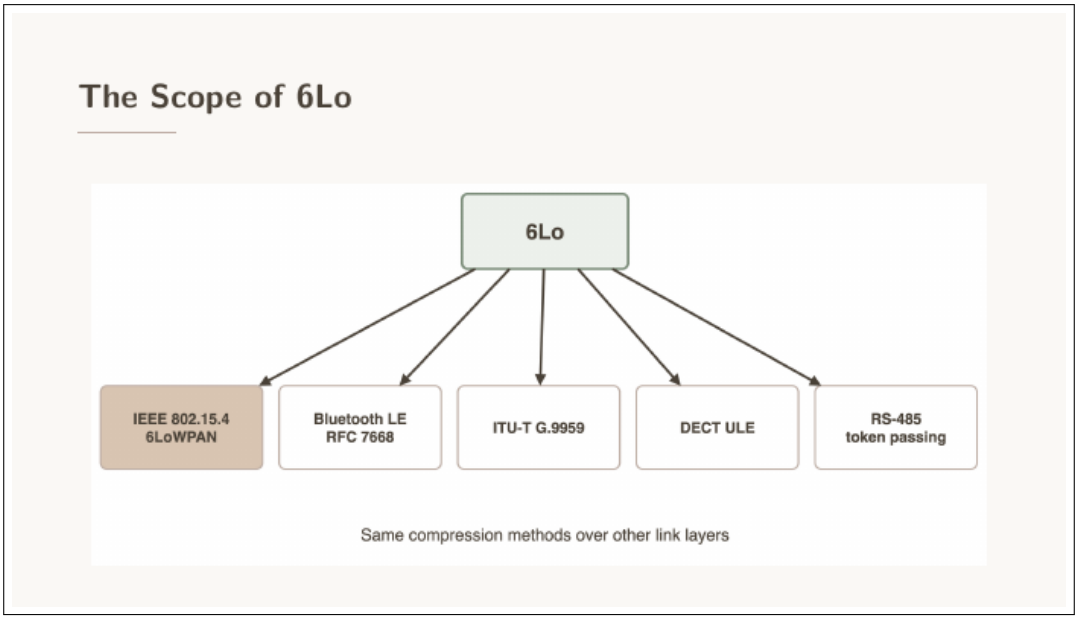
The end of 6LoWPAN and the start of 6Lo. The 6LoWPAN working group finished its work in 2012. 6Lo was created in 2013 to reuse the same mechanisms on other link technologies, which shows that the idea generalised.

Slide 27. From 6LoWPAN to 6Lo (2/2)



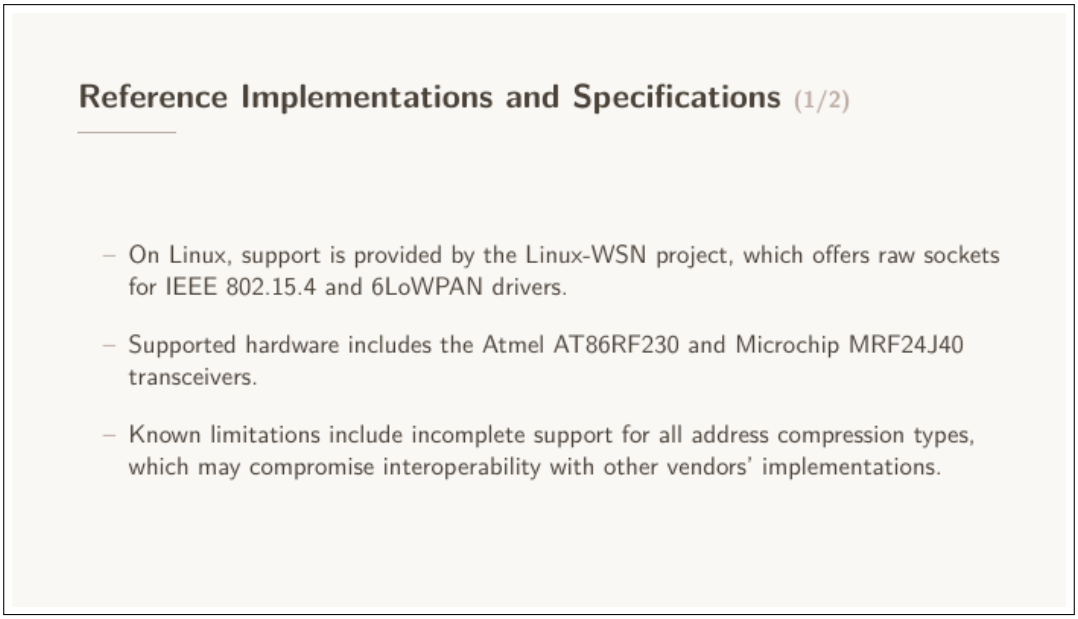
The Bluetooth Low Energy case. On BLE, RFC 7668 dispenses with fragmentation, because L2CAP already fragments, but keeps the compression methods. This links the lecture to lectures 4 to 6, where BLE was studied without IP.

Slide 28. The Scope of 6Lo



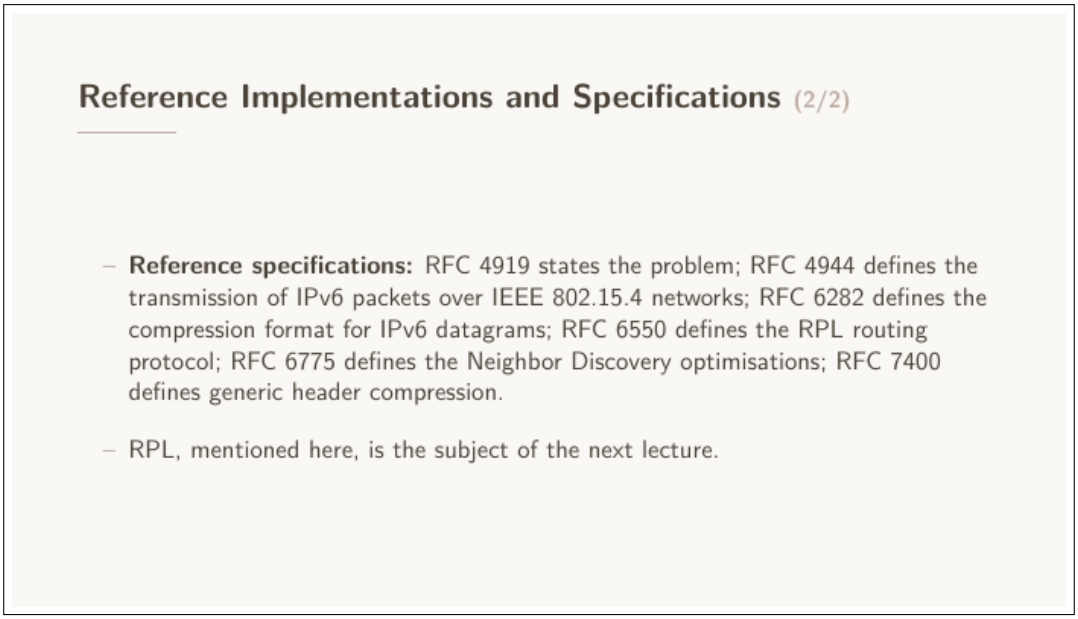
Same methods, other links. The figure shows 6Lo as an umbrella over 802.15.4, BLE, G.9959, DECT ULE and RS-485. In all of them, the compression methods are reused.

Slide 29. Reference Implementations and Specifications (1/2)



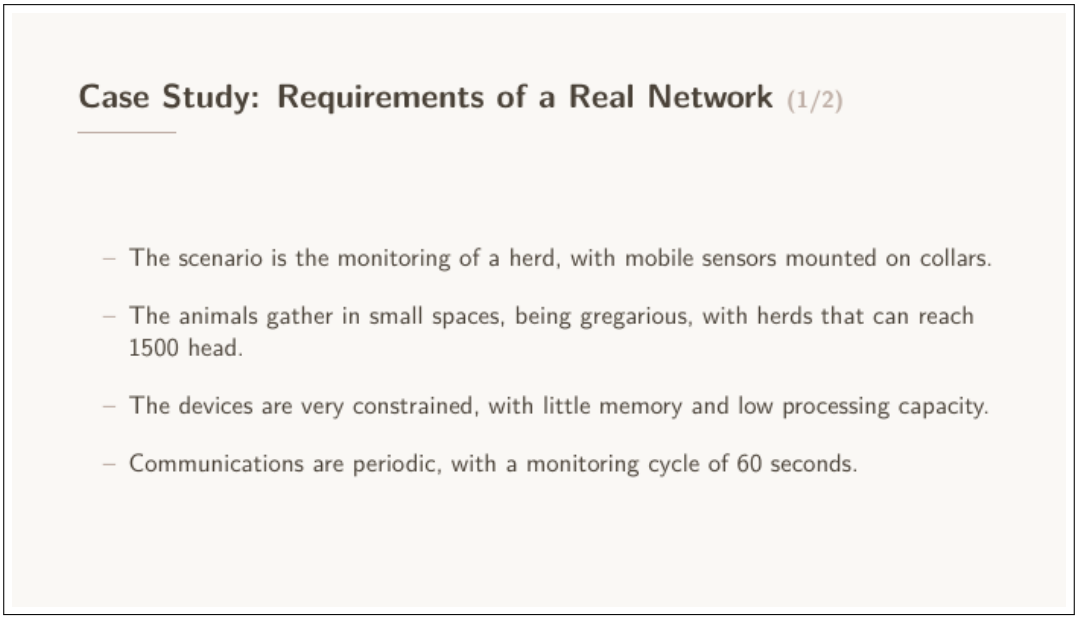
The state of the implementation. Linux support exists, but does not cover all compression types. The warning is practical: interoperability between implementations from different vendors should be tested, not assumed.

Slide 30. Reference Implementations and Specifications (2/2)



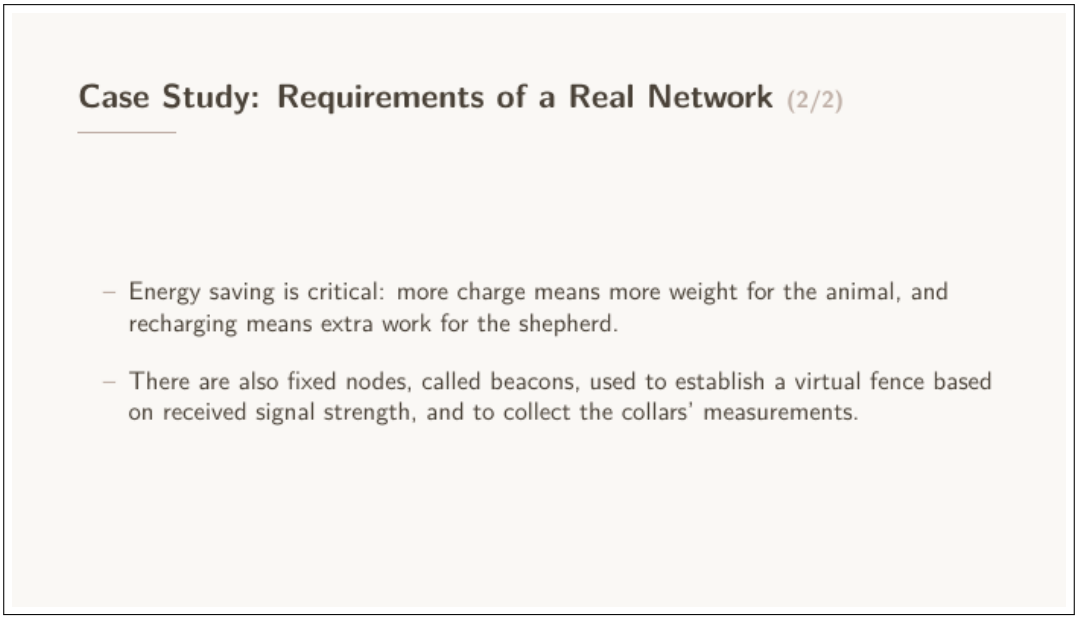
The normative references. The RFCs listed are the basic bibliography. You should be able to associate each with its topic: 4944 with transmission, 6282 with compression, 6775 with Neighbor Discovery, 7400 with generic compression, 6550 with RPL.

Slide 31. Case Study: Requirements of a Real Network (1/2)



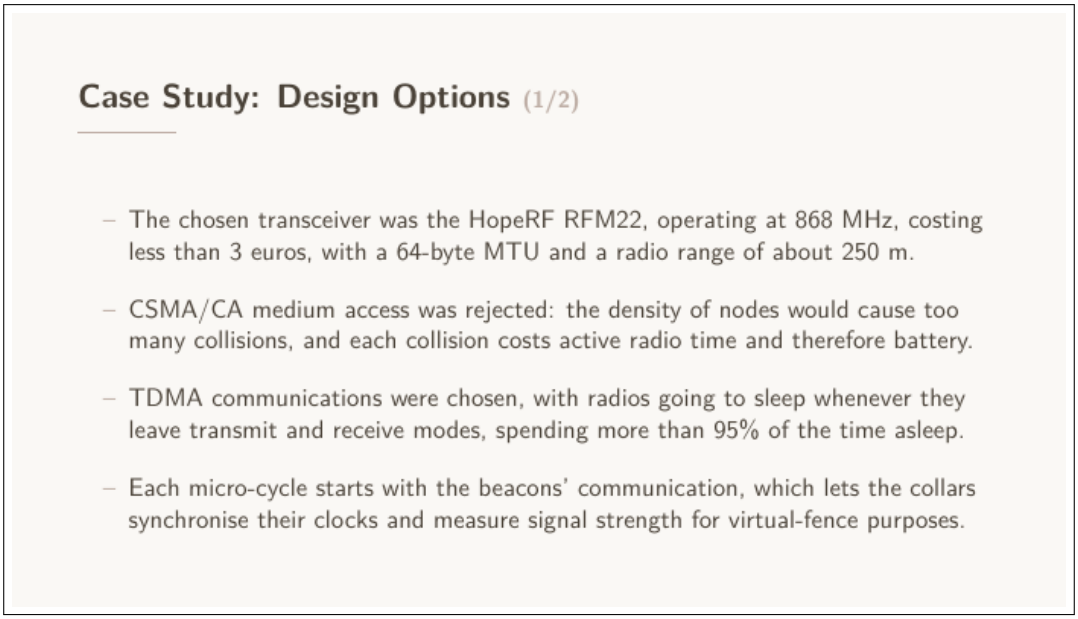
The requirements of the case. Monitoring a herd imposes many nodes, high density, a 60-second cycle and very constrained devices. Each requirement weighs on the decisions the next slide describes.

Slide 32. Case Study: Requirements of a Real Network (2/2)



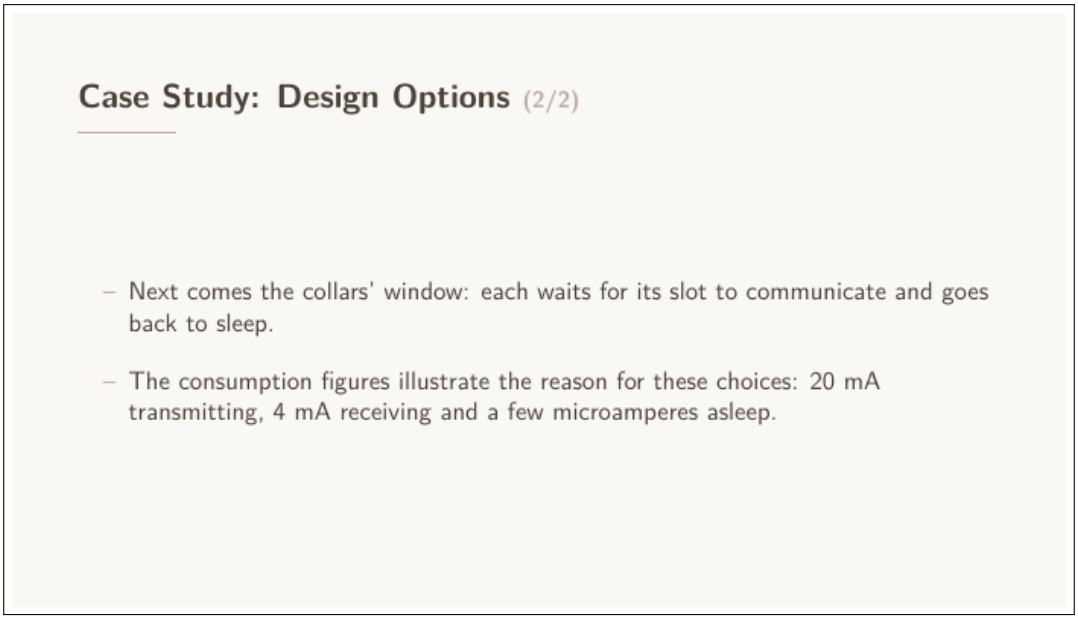
Energy and infrastructure. The weight of the battery and the work of recharging make energy the dominant requirement. The fixed beacons serve as a reference for the virtual fence and to collect the data.

Slide 33. Case Study: Design Options (1/2)



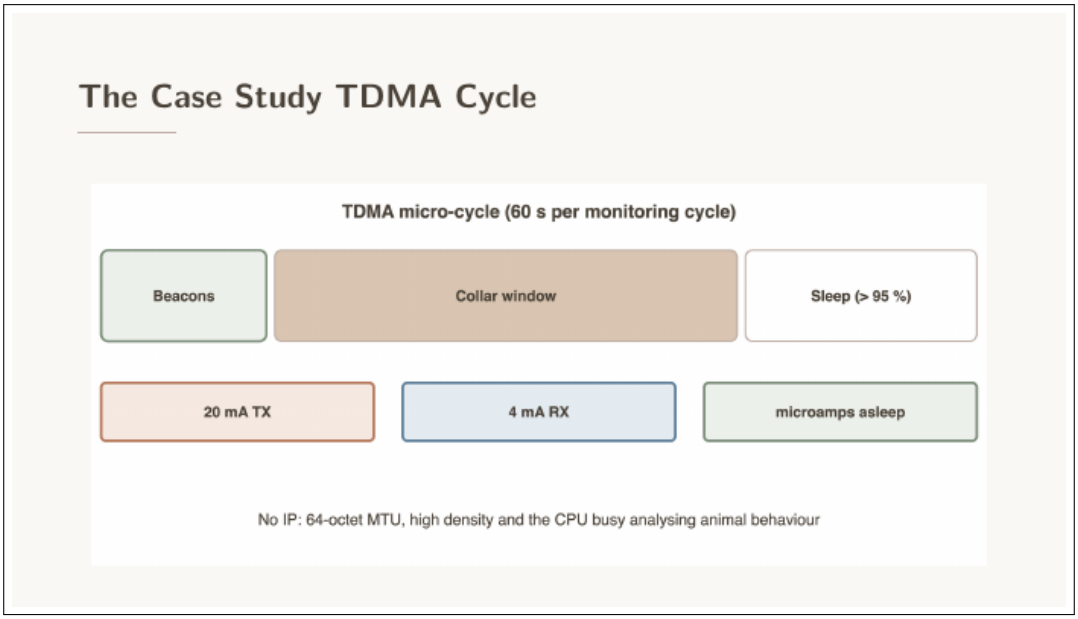
The design choices. The low-cost radio, TDMA instead of CSMA/CA and sleeping more than 95

Slide 34. Case Study: Design Options (2/2)



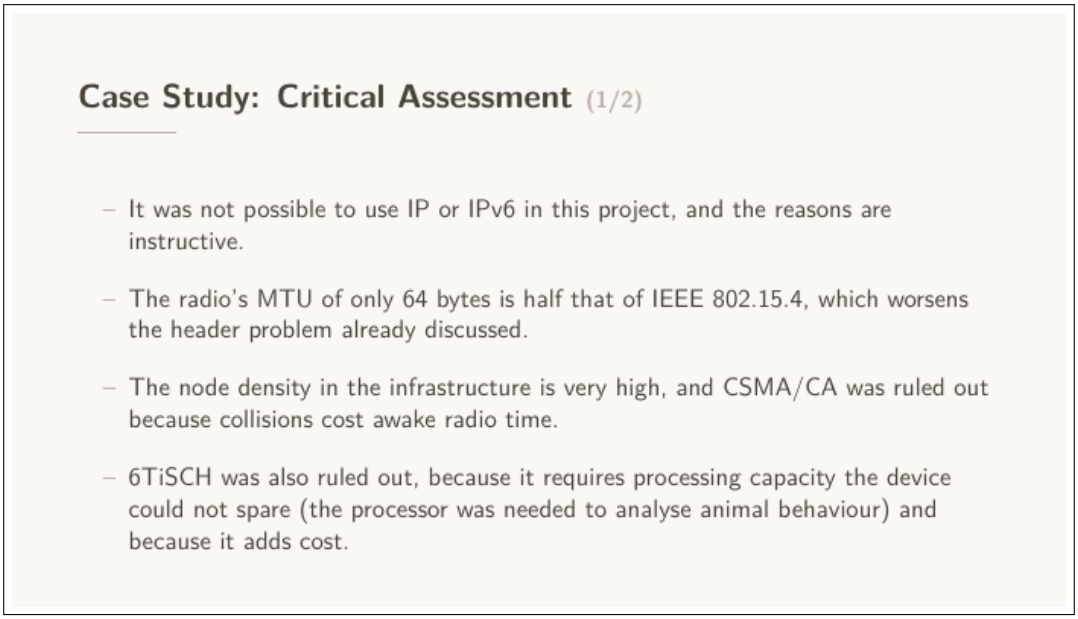
Consumption explains the options. 20 mA transmitting, 4 mA receiving and microamps asleep. The difference in orders of magnitude explains why most of the project consists of keeping the radio off.

Slide 35. The Case Study TDMA Cycle



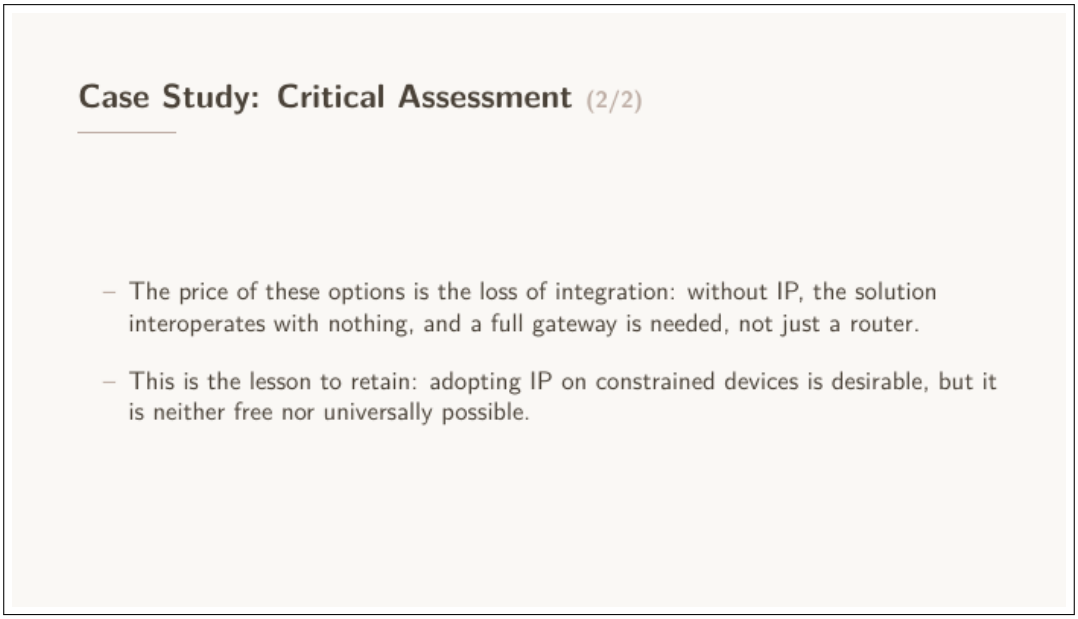
The TDMA micro-cycle. The figure shows the three phases: beacons, collar window and sleep. Synchronisation is done by the beacons, and each collar wakes up only at its instant. Sleep takes up most of the cycle.

Slide 36. Case Study: Critical Assessment (1/2)



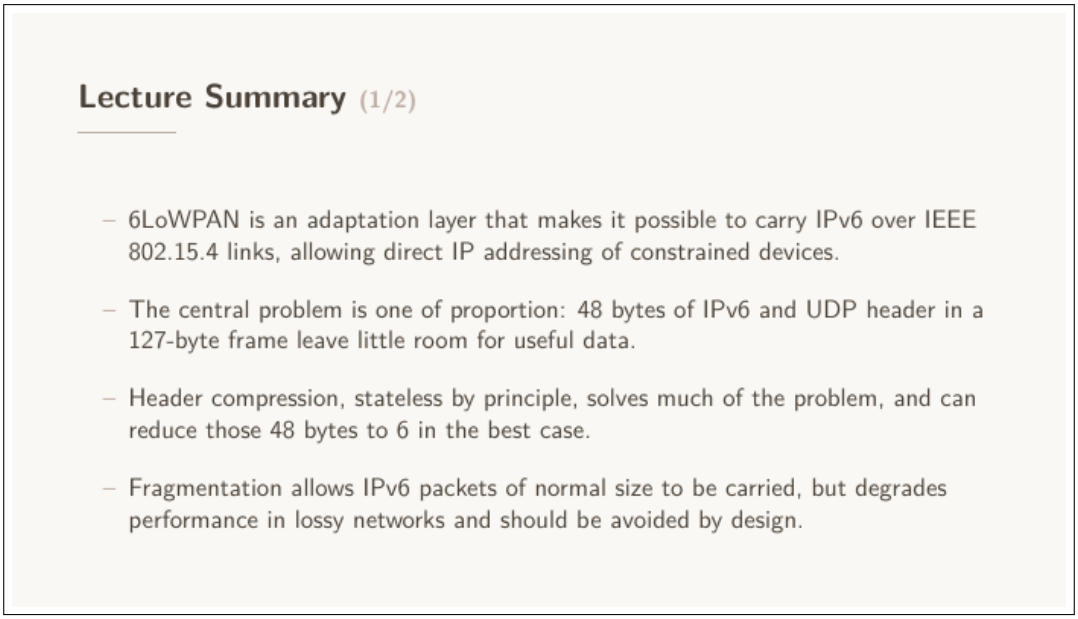
Why IP was not used. The 64-octet MTU, the node density and the lack of processing capacity ruled out IP, CSMA/CA and 6TiSCH. Read this slide as an exercise in justifying decisions.

Slide 37. Case Study: Critical Assessment (2/2)



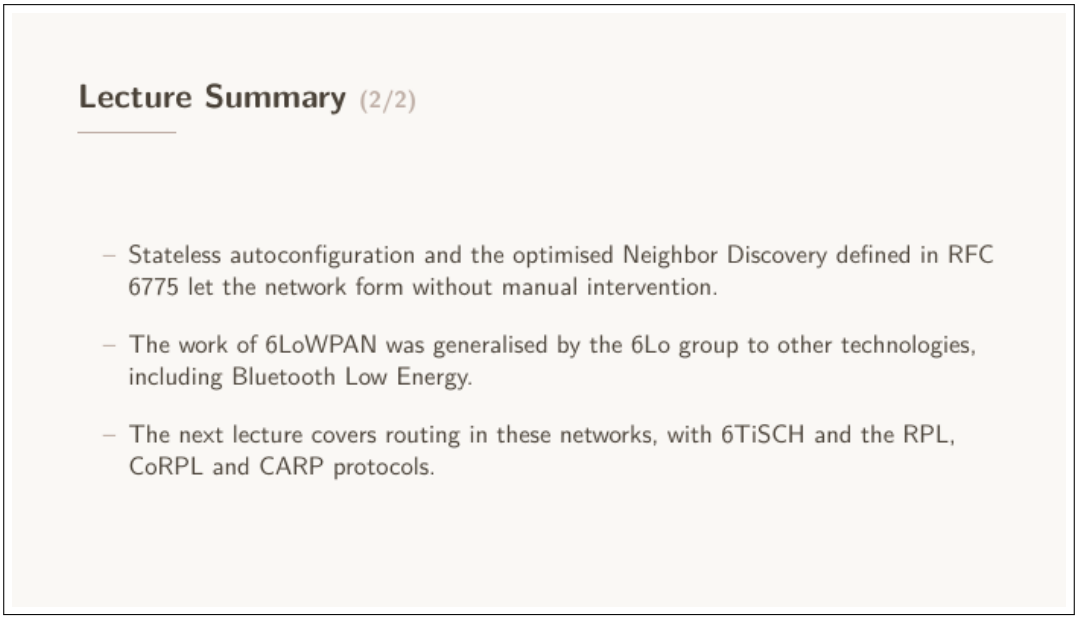
The price of the decision. Without IP, interoperability is lost and a full gateway is needed. The lesson is that IP is desirable but not free, and that the decision should be taken with the numbers in front of you.

Slide 38. Lecture Summary (1/2)



The essentials of 6LoWPAN. Adaptation layer, a problem of proportion, stateless compression and fragmentation to be avoided. The number to retain is the reduction from 48 to 6 octets in the best case.

Slide 39. Lecture Summary (2/2)



What comes next. Autoconfiguration and optimised Neighbor Discovery complete the base, and 6Lo extends it to other technologies. The next lecture covers routing, which is what is still missing for these networks to work end to end.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. How many payload octets are left in an 802.15.4 frame with a 25-octet MAC header, IPv6 and UDP?
2. Why is stateless compression important on constrained devices?
3. What is the dispatch code and what is it for?
4. Why should fragmentation be avoided in lossy networks?
5. How is the interface identifier built from the 16-bit short address?
6. Under what conditions is it justified not to use IP in an IoT project?

References

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